

SHAOQING DAI



WORKING EXPERIENCE

Current
|
2025.01

Key-Funded Postdoctoral Researcher

Geography Postdoctoral Program, School of Resource and Environmental Sciences, Wuhan University

📍 Wuhan, China

• Co-mentor: Prof.Peng Jia



EDUCATION

2024.06
|
2020.01

Ph.D., Geoinformatics

Faculty of Geo-information Science and Earth Observation(ITC), University of Twente

📍 Enschede, Netherlands

- Thesis: Improving obesogenic environmental assessments with advanced geospatial methods
- Supervised by Prof.Peng Jia, Prof.Alfred Stein

2019.06
|
2016.09

M.Sc., Ecology

Institute of Urban Environment, Chinese Academy of Sciences, University of Chinese Academy of Sciences

📍 Xiamen, China

- Thesis:Creating a Carbon Source and Sink Map by Coupling an Ecological Process Model with an Emission Inventory to Study a Carbon Balance
- Supervised by Prof.Yin Ren

2016.06
|
2012.09

B.S., Geographical Science (National Talents Training Base)

School of Geographical Sciences, Fujian Normal University 📍 Fuzhou, China

- Thesis:The Planning Location and Design of Smart Campus Based on BIM and GIS:A Case Study of Fujian Normal University
- Supervised by Prof.Huixian Jiang



RESEARCH INTERESTS

Current
|
2012

Health Geography; Urban Visual Intelligence; Spatial Lifecourse Helath; Spatial Statistics; GeoAI; Spatial-Temporal Big Data



View this CV online with links at nickstrayer.me/cv/

CONTACT

✉ shaoqing.dai@outlook.com

🌐 [GISerDaiShaoqing](#)

🏠 [Homepage](#)

LANGUAGE SKILLS

R	
GIS	
3D Modeling	
Python	
Statistics	
RS & Earth Observation	

Made with the R package [pagedown](#).

The source code is available on [GISerDaiShaoqing/Curriculum-Vitae-Latex](#).

Last updated on 2026-01-28.



DEVELOPED OPEN SOFTWARE

Current
|
2017

• **Note of Applied Statistics with R**

- <http://gisersqdai.top/Note-of-Applied-Statistics-with-R-Book/>

Current
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2018

• **D3L Tool of NASA Satellite**

- The program provides a graphical interface to download satellite imagery from NASA, such as MODIS products, MERIS and more information can see url¹

Current
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2018

• **Geographical detector in R**

- The geographical detector software based on R

Current
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2018

• **GitHub Paper Notebook**

- The software for notebook of papers

Current
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2019

• **rgeoda**

- rgeoda is a R package for spatial data analysis based on libgeoda and GeoDa(contributor).

Current
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2020

• **COVID-19 Dashboard shiny app**

- A developed dashboard shiny app about COVID-19, url².

Current
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2020

• **Awesome COVID-19**

- Resources of COVID-19 research. <http://covid19.gisersqdai.top/en-us/>

Current
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2023

• **MetS score calculator shiny app**

- A shiny app to calculate MetS score. https://gisersqdai.shinyapps.io/mets_severiity_calculator/



AWARDS & HONORS

2025

• **Best Oral Presentation Award in 3rd Urban Agglomeration Green Development Conference**

- Urban Agglomeration Green Development Special Committee, Chinese Society for Urban Studies

2025	The National 2nd Prize in the 9th Planning Decision Support Model Design Contest³Supervisor • Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd.
2025	Special Program for the Introduction of Overseas Postdoctoral Talents • Ministry of Education of the People's Republic of China
2022	Outstanding contributions to COVID-19 epidemic prevention and control • Health commission of Tumot Left Banner, Inner Mongolia autonomous region
2021	The National 2nd Prize in the 5th Planning Decision Support Model Design Contest(Chengyuan Cup) • Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd., Beijing City Lab, Baidu Map Huiyan, China Unicom Smart Footprint
2019	China Scholarship Council(CSC) • China Scholarship Council
2019	The 2nd Grade Scholarship of Yonker Environmental Protection Scholarship • Yonker Environmental Protection Co. Ltd.
2019	The 1st Grade Scholarship of Institute of Urban Environment, Chinese Academy of Sciences • Institute of Urban Environment, Chinese Academy of Sciences
2019	The Scientific and Technological Innovation Award of Outstanding Graduate Students of the Ecological Society of China • Ecological Society of China
2018	National Scholarship for M.Sctop 2-3% • Ministry of Education of the People's Republic of China

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- 2018 • **The National 3rd Prize in the 2nd Big Data Support Spatial Planning and Design Competition**
- Urban Big Data Special Committee, Chinese Society for Urban Studies
- 2018 • **“Applied Energy”2018 Summer School Certification**
- The Applied Energy Innovation Institute, Department of Earth System Science, Tsinghua University, China Emission Accounts & Datasets(CEADs)
- 2018 • **Merit Students in the academic year 2017-2018**
- University of Chinese Academy of Sciences
- 2018 • **The 2nd Grade Scholarship of Institute of Urban Environment, Chinese Academy of Sciences**
- Institute of Urban Environment, Chinese Academy of Sciences
- 2018 • **The 1st Scholarship of Institute of Urban Environment, Chinese Academy of Sciences**
- Institute of Urban Environment, Chinese Academy of Sciences
- 2018 • **The National Winning Award in 2017 “Sharing Cup” College Student Science and Technology Resource Sharing Service Innovation Contest**
- National Science & Technology Infrastructure Center
- 2017 • **“Excellent Volunteer” award of 15th User Conference of Esri China**
- Esri China Information Technology Co. Ltd.
- 2017 • **Merit Students in the academic year 2016-2017**
- University of Chinese Academy of Sciences
- 2015 • **The National Winning Award in 2015 Esri Cup National Collegiate Developed Contest in GIS-3D Modeling and Design**
- Chinese Society of Geodesy Photogrammetry and Cartography, Esri China Information Technology Co. Ltd.
- 2015 • **Inovation Scholoarships in the academic year 2014-2015**
- Fujian Normal University

2015 • **The National 3rd Prize in the Smart City Special Event Contest, the 14th Challenge Cup National College Students' Extracurricular Academic Science and Technology Works Contest**

• Chinese Communist Youth League, China Association for Science and Technology, Ministry of Education of the People's Republic of China, All-China Students' Federation

2015 • **Scientific and Technological Innovation award and Nomination, the First Annual Fujian Normal University Student Selection**

• Fujian Normal University

2015 • **The School Wining Award during the 9th Challenge Cup National College Students' Extracurricular Academic Science and Technology Works Contest**

• Fujian Normal University

2014 • **Inovation Scholoarships in the academic year 2013-2014**

• Fujian Normal University

2014 • **The National 3rd Prize in 2014 MathorCup Global Collegiate Mathematical Contest in Modeling**

• Chinese Society of Optimization, Overall Planning and Economical Mathematics

2014 • **"Excellent League Cadres" in the academic year 2013-2014**

• Fujian Normal University

2014 • **"Excellent League Cadres" in the academic year 2013-2014**

• School of Geographical Sciences, Fujian Normal University

2013 • **"Excellent League Cadres" in the academic year 2012-2013**

• School of Geographical Sciences, Fujian Normal University



ACADEMIC SERVICE & AFFILIATIONS

Current
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2025

Editors

BMC Pbulic Health

• Editor Board Member

Current 2025	Editors Frontiers in Public Health, Frontiers in Psychology • Associate Editor
Current 2025	Editors Land Special Issue, "Urban Spatial Planning for Health and Well-Being" • Guest Editor
Current 2025	Editors Frontiers in Remote Sensing Special Issue, "Machine Learning for Advanced Remote Sensing: From Theory to Applications and Societal Impact" • Guest Editor
Current 2024	Affiliations AI 4D City lab, Hong Kong University of Science and Technology (Guangzhou) • Visiting Scholar
Current 2019	Affiliations International Institute of Spatial Lifecourse Health (ISLE) • Fellow
Current 2019	Affiliations GeoDa Center, rgeoda, libgeoda Github Organizations • Contributor
Current 2015	Soical Media Volunteer The Organizations "Sustainable Cities & Mobility" • Volunteer
Current 2025	Conference & Society Service The Geographical Society of China • Member
Current 2024	Conference & Society Service International Society for Photogrammetry and Remote Sensing Student Consortium • Member
Current 2024	Workshop Lecturer UrbanParty • Lecturer

Current
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2019

• **Peer-reviewer Journal & Conference Reviewer**

European Heart Journal, Resources, Conservation and Recycling, Applied Energy, Sustainable Cities and Society, International Journal of Geographical Information Science, Habitat International, Cities, Information Processing & Management, Ecological Indicators, Methods in Ecology and Evolution, International Journal of Digital Earth, GIScience & Remote Sensing, Information Geography, Forest Ecology & Management, Scientific Data, Carbon Management, City and environment interactions, Transactions in Urban Data, Science, and Technology, Frontiers in Public Health, Frontiers in Environmental Science, Frontiers in Earth Science, Scientific Reports, Data in Brief, Applied Economic Letters, Children, HealthCare, The Journal of Open Source Software, Urban & BuildingScience, Shanghai Urban Planning Review, IPSRS Conference, CPGIS Conference, Biology and Life Sciences Forum

• Reviewer

2025

• **Science & Technology Service**

The 138th Postdoctoral Science and Technology Service Corps, China Postdoctoral Science Foundation

• Fellow

2025

• **Conference & Society Service**

2nd National Conference of Information Geography, "Human behaviors, health geography and spatial modeling" session

• Convener

2025

• **Conference & Society Service**

5th International Symposium on Spatial Lifecourse Health, "Street view images, human mobility and environmental health" session

• Convener

2025

• **Workshop Lecturer**

5th International Symposium on Spatial Lifecourse Health Workshop

• Lecturer

2024

• **Conference & Society Service**

2nd World Congress of Geography, "Research on Global Health Wellbeing and Health Geography" session

• Convener

2024

• **Conference & Society Service**

4th International Symposium on Spatial Lifecourse Health, "Application of System Simulation Models in Health Domain" session

• Convener

- 2024 • **Workshop Lecturer**
4th International Symposium on Spatial Lifecourse Health Workshop
• Lecturer
- 2024 • **Workshop Lecturer**
Heywhale Community and GISphere, 'Python x GIS' GeoAI Workshop
• Lecturer
- 2024 • **Workshop Lecturer**
Training course on 'Spatial Epidemiology Analysis Methods and Applications'
• Lecturer
- 2023
|
2020 • **Soical Media Volunteer**
GISphere
• Editor Board, Council Member, Editor in chief of Wechat
- 2020
|
2018 • **Science & Technology Service**
Aliyunqi Community
• Expert
- 2019
|
2017 • **Conference & Society Service**
International Association of Landscape Ecology of China(IALE-China)
• Member
- 2017
|
2013 • **Editors**
The Geographical Tribune of Youth
• Founder, Editor in Chief, Associate Editor, Assistant Editor in Chief
- 2017
|
2016 • **Soical Media Volunteer**
Green Bike-Transit
• Volunteer



PUBLICATIONS-PEER-REVIEWED JOURNAL ARTICLES

- 2025 • **Effects of short-term exposure to air pollutants on real-time blood pressure: a wearable device-based study in China**
Journal of Urban Health
• Chen Li, Xiu Yang, Kun Qin, Shuhan Yang, Shaoqing Dai, Chun Yin, Shujuan Yang, Yuanyuan Shi, Peng Jia
• 102(5): 1011-1023. [doi:10.1007/s11524-025-01017-3](https://doi.org/10.1007/s11524-025-01017-3).
• SCI, IF=4.1

- 2025 • **Assessing variations in river networks under urbanization across metropolitan plains using a multi-metric approach**
Land
• Zhixin Lin, Shuang Luo, Miao Lu, Shaoqing Dai, Youpeng Xu
• 14(10): 1994. doi:10.3390/land14101994.
• SSCI, IF=3.2
- 2025 • **Agricultural socialized service system and its mechanism driving industrial upgrade: based on an empirical study on value chain theory**
Resources & Industries
• Yuxei Lin, Feng Lu, Zhengyi Xu, Junfeng Xue, Shaoqing Dai, Xinying Zhang
• 7(05): 139-150. doi:10.13776/j.cnki.resourcesindustries.20250901.006.
• in Chinese
- 2025 • **High-resolution population density mapping in urban areas using a contextualized geographically weighted neural network (CGWNN) model**
Applied Geography
• Ge Qiu, Yuchen Li, Kun Qin, Chen Li, Shuhan Yang, Chun Yin, Yaolin Liu*, Shaoqing Dai*, Peng Jia*
• 182: 103708. doi:10.1016/j.apgeog.2025.103708.
• SSCI, IF=5.4
- 2025 • **Identifying landscape patterns at different scales as driving factors for urban flooding**
Ecological Indicators
• Yao Li, Frank Badu Osei, Shaoqing Dai, Tangao Hu, Alfred Stein.
• 176: 113614. doi:10.1016/j.ecolind.2025.113614.
• SCI, IF=7.4
- 2025 • **Built environments and obesity: A framework considering residences, commute routes, and workplaces**
Cities
• Chun Yin, Zhe Zhang, Shaoqing Dai, Yiyi Chen
• 160: 105842. doi:10.1016/j.cities.2025.105842.
• SSCI, IF=6.6
- 2025 • **Toward 3D hedonic price model for vertically developed cities using street view images and machine learning methods**
Habitat International
• Yue Ying, Shaoqing Dai*, Mila Koeva, Monika Kuffer, Claudio Persello, Wen Zhou, Jaap Zevenbergen
• 156: 103288. doi:10.1016/j.habitatint.2025.103288.
• SSCI, IF=7.0

- 2025 • **Heterogeneous impacts of and vulnerabilities to the COVID-19 pandemic**
Landscape Ecology
- Manyao Li#, Shaoqing Dai#, Yuanyuan Shi#, Kun Qin#, Ross Brownson, Yan Kestens, Miyang Luo, Shiyong Liu, Jing Su, Gordon G. Liu, Shujuan Yang, Peng Jia
 - 40: 32. doi:10.1007/s10980-024-02039-z
 - SCI, IF=3.7
- 2024 • **Long-term exposure to PM2.5 chemical constituents and diabetes: evidence from a multi-center cohort study in China**
The Lancet Regional Health–Western Pacific
- Changwei Cai, Shuzhen Zhu, Mingfang Qin, Xiaoqing Li, Chuanteng Feng, Bin Yu, Shaoqing Dai, Ge Qiu, Yuchen Li, Tingting Ye, Wenling Zhong, Ying Shao, Lan Zhang, Peng Jia, Shujuan Yang
 - 47: 101100. doi:10.1016/j.lanwpc.2024.101100
 - SCI, IF=8.1
- 2024 • **Street view imagery-based built environment auditing tools: a systematic review**
International Journal of Geographical Information Science
- Shaoqing Dai, Yuchen Li, Alfred Stein, Shujuan Yang, Peng Jia.
 - 38(6): 1136–1157. doi:10.1080/13658816.2024.2336034
 - SCI/SSCI, IF=5.1
- 2024 • **Spatiotemporal patterns of air pollutants over the epidemic course: a national study in China**
Remote Sensing
- Kun Qin#, Zhanpeng Wang#, Shaoqing Dai#, Yuchen Li, Manyao Li, Chen Li, Ge Qiu, Yuanyuan Shi, Chun Yin, Shujuan Yang, Peng Jia*
 - 16(7): 1298. doi:10.3390/rs16071298.
 - SCI, IF=4.1
- 2024 • **State of the art of lifecourse cohort establishment**
China CDC Weekly
- Shaoqing Dai, Ge Qiu, Yuchen Li, Shuhan Yang, Shujuan Yang, Peng Jia
 - 6(14), 300–304. doi:10.46234/ccdcw2024.058.
 - SCI/SSCI, IF = 2.9
- 2024 • **Long-term dynamic monitoring and driving force analysis of eco-environmental quality in China**
Remote Sensing
- Weiwei Zhang, Zixi Liu, Kun Qin, Shaoqing Dai, Huiyuan Lu, Miao Lu, Jianwan Ji, Zhaohui Yang, Chao Chen, Peng Jia
 - 16(6): 1028 doi:10.3390/rs16061028.
 - SCI, IF=4.1

- 2024 • **Mixed land use has opposite associations with subjective well-being through social capital: Spatial heterogeneity in residential and workplace neighborhoods**
Travel Behaviour and Society
• Chun Yin, Jiahang Liu, Shaoqing Dai, Bindong Sun
• 35, 100724. doi:10.1016/j.tbs.2023.100724.
• SSCI, IF=5.7
- 2024 • **Local carbon emission zone construction in the highly urbanized regions: Application of residential and transport CO₂ emissions in Shanghai, China**
Building and Environment
• Yunfeng Tian, Shudi Zuo, Jiaheng Ju, Shaoqing Dai, Yin Ren, Panfeng Dou.
• 247, 111007. doi:10.1016/j.buildenv.2023.111007.
• SCI, IF=7.6
- 2023 • **Assessing Spatiotemporal Bikeability Using Multi-Source Geospatial Big Data: A Case Study of Xiamen, China**
International Journal of Applied Earth Observation and Geoinformation
• Shaoqing Dai, Wufan Zhao, Yanwen Wang, Xiao Huang, Zhidong Chen, Jinghan Lei, Alfred Stein, Peng Jia.
• 125, 103539. doi:10.1016/j.jag.2023.103539.
• SCI, IF=8.6
- 2023 • **Development and validation of an age-sex-ethnicity-specific metabolic syndrome score in the Chinese adults**
Nature Communications
• Shujuan Yang, Bin Yu, Wanqi Yu, Shaoqing Dai, Chuanteng Feng, Ying Shao, Xing Zhao, Xiaoqing Li, Tianjing He, Peng Jia.
• 14(1), 6988. doi:10.1038/s41467-023-42423-y
• SCI, IF=15.7
- 2023 • **Environmental determinants of childhood obesity: a meta-analysis**
The Lancet Global Health
• Peng Jia, Yuanyuan Shi, Qi Jiang, Shaoqing Dai, Bin Yu, Shuhan Yang, Ge Qiu, Shujuan Yang.
• 11, S7. doi:10.1016/S2214-109X(23)00092-X.
• SCI, IF=18.0
- 2022 • **The importance of the functional mixed entropy for the explanation of residential and transport CO₂ emissions in the urban center of China**
journal of Cleaner Production
• Shudi Zuo, Shaoqing Dai, Jiaheng Ju, Fanxin Meng, Yin Ren, Yunfeng Tian, Kaide Wang
• 380, 134947. doi:10.1016/j.jclepro.2022.134947.
• SCI, IF=10.0

- 2022 • **Global spreading of Omicron variant of COVID-19**
Geospatial Health
- Pengxin Zhang, Shuhan Yang, Shaoqing Dai, Darren How Jin Aik, Shujuan Yang, Peng Jia.
 - Geospatial Health. 17. doi:10.4081/gh.2022.1083.
 - SCI, IF=0.9
- 2022 • **Quantifying urban mass gain and loss by a GIS-based material stocks and flows analysis. Journal of Industrial Ecology**
Journal of Industrial Ecology
- Yupeng Liu, Jiajia Li, Weiqiang Chen, Lulu Song, Shaoqing Dai.
 - 26(3), 1051-1060. doi:10.1111/jiec.13252.
 - SCI, IF=5.4
- 2022 • **Identification of two-dimensional copper signatures in human blood for bladder cancer with machine learning**
Chemical Science
- Weichao Wang, Xian Liu, Changwen Zhang, Fei Sheng, Shanjun Song, Penghui Li, Shaoqing Dai, Bin Wang, Dawei Lu, Luyao Zhang, Xuezhi Yang, Zhihong Zhang, Sijin Liu, Aiqian Zhang, Qian Liu, Guibin Jiang.
 - 3(6), 1648-1656. doi:10.1039/D1SC06156A.
 - SCI, IF=7.4
- 2021 • **Improving plot-level model of forest biomass: A combined approach using machine learning with spatial statistics**
Forests
- Shaoqing Dai, Xiaoman Zheng, Lei Gao, Chengdong Xu, Shudi Zuo, Qi Chen, Xiaohua Wei, Yin Ren
 - 12(12), 1663. doi:10.3390/f12121663.
 - SCI, IF=2.5
- 2021 • **New approaches to anticipate the risk of reverse zoonosis**
Trends in Ecology & Evolution
- Peng Jia, Shaoqing Dai, Tong Wu, Shujuan Yang.
 - 36(7), 580-590. doi:10.1016/j.tree.2021.03.012.
 - SCI, IF=17.3
- 2021 • **Refined water security assessment for sustainable water management: A case study of 15 key cities in the Yangtze River Delta, China**
Journal of Environmental Management
- Panfeng Dou, Shudi Zuo, Yin Ren, Manuel J. Rodriguez, Shaoqing Dai.
 - 290, 112588. doi:10.1016/j.jenvman.2021.112588.
 - SCI, IF=8.4

- 2021 • **Pitfalls in Modeling Asymptomatic COVID-19 Infection**
Frontiers in Public Health
- Shujuan Yang, Shaoqing Dai, Yulin Huang, Peng Jia.
 - 9, 593176. doi:10.3389/fpubh.2021.593176.
 - SCI, IF=3.4
- 2021 • **High spatial resolution mapping of steel resources accumulated above ground in mainland China: Past trends and future prospects**
Journal of Cleaner Production
- Lulu Song, Shaoqing Dai, Zhi Cao, Yupeng Liu, Weiqiang Chen.
 - 297, 144609. doi:10.1016/j.jclepro.2021.126482.
 - SCI, IF=10.0
- 2020 • **The direct and interactive impacts of hydrological factors on bacillary dysentery across different geographical regions in central China.**
Science of the Total Environment
- Shudi Zuo, Lianping Yang, Panfeng Dou, Hung Chak Ho, SQ. Dai, Wenjun Ma, Yin Ren, Cunrui Huang.
 - 764, 144609. doi:10.1016/j.scitotenv.2020.144609.
 - SCI, IF=8.0
- 2020 • **Investigating the uncertainties propagation analysis of CO₂ emissions emissions gridded maps at the urban scales: a case study of Jinjiang city, China.**
Remote Sensing
- Shaoqing Dai, Yin Ren, Shudi Zuo, Chengyi Lai, Jiajia Li, Shengyu Xie, Bingchu Chen.
 - 12(23), 3932. doi:10.3390/rs12233932
 - SCI, IF=4.1
- 2020 • **The Evaluation of Health Effect of Short-term Exposure to PM_{2.5} during Spring Festival: A Case Study of 25 Cities in the Yangtze River Delta**
Shanghai Urban Planning Review
- Shaoqing Dai, Jiajia Li, Weixu Yang, Fangyu Chen, Huixian Jiang.
 - 05, 22-29. doi:10.11982/j.supr.20200504.
 - in Chinese
- 2020 • **Natural environment and childhood obesity: A systematic review**
Obesity Reviews
- Peng Jia, Shaoqing Dai, Kristen. E. Rohli, Robert V. Rohli, Yanan Ma, Chao Yu, Xiongfeng Pan, Weiqi Zhou.
 - 1-9. doi:10.1111/obr.13097
 - SCI, IF=7.4

- 2020 • **Response of syntrophic bacterial and methanogenic archaeal communities in paddy soil to soil type and phenological period of rice growth**
Journal of Cleaner Production
- XiaoFang Pan, Hu Li, Lixin Zhao, Xiaoru Yang, Jianqiang Su, Shaoqing Dai, Jing Ning, Chunxing Li, Guanqing Cai, GeFu Zhu.
 - 278, 123418. doi:10.1016/j.jclepro.2020.123418
 - SCI, IF=10.0
- 2020 • **Green space access in the neighbourhood and childhood obesity**
Obesity Reviews
- Peng Jia, Xinxi Cao, Hongxi Yang, Shaoqing Dai, Pan He, Ganlin Huang, Tong Wu, Yaogang Wang.
 - 1-12. doi:10.1111/obr.13100
 - SCI, IF=7.4
- 2020 • **Neighbourhood speed limit and childhood obesity**
Obesity Reviews
- Miyang Luo, Hanqi Li, Xiongfen Pan, Teng Fei, Shaoqing Dai, Ge Qiu, Yuxuan Zou, Heleen Vos, Jiayou Luo, Peng Jia.
 - 1-11. doi:10.1111/obr.13052
 - SCI, IF=7.4
- 2020 • **Treatment of high-ash industrial sludge for producing improved char with low heavy metal toxicity**
Journal of Analytical and Applied Pyrolysis
- Shengyu Xie, Guangwei Yu, Chunxing Li, Jie Li, Gang Wang, Shaoqing Dai, Yin Wang.
 - 150, 104866. doi:10.1016/j.jaap.2020.104866
 - SCI, IF=6.2
- 2020 • **Access to bike lanes and childhood obesity: A systematic review and meta-analysis**
Obesity Reviews
- Xiongfen Pan, Li Zhao, Jiayou Luo, Yinhao Li, Lin Zhang, Tong Wu, Melody Smith, Shaoqing Dai, Peng Jia.
 - 1-11. doi:10.1111/obr.13042
 - SCI, IF=7.4
- 2020 • **A spatial database of CO₂ emissions, urban form fragmentation and city-scale effect related impact factors for the low carbon urban system in Jinjiang city, China**
Data in Brief
- Shaoqing Dai, Shudi Zuo, Yin Ren.
 - 29, 105274. doi:10.1016/j.dib.2020.105274
 - ESCI, IF=1.4

- 2020 • **More fragmentized urban form more CO₂ emissions? An comprehensive relationship from the combination analysis across different scales**
Journal of Cleaner Production
- Shudi Zuo, Shaoqing Dai, Yin Ren.
 - 244, 118659. [doi:10.1016/j.jclepro.2019.118659](https://doi.org/10.1016/j.jclepro.2019.118659)
 - SCI, IF=10.0
- 2019 • **The impacts of climate and land use/land cover changes on water yield service in Ningbo region**
Acta Scientiae Circumstantiae
- Panfeng Dou, Shudi Zuo, Yin Ren, Shaoqing Dai, Guoliang Yun
 - 39(7), 2398-2409. [doi:10.13671/j.hjkxxb.2019.0122](https://doi.org/10.13671/j.hjkxxb.2019.0122)
 - CSCD, in Chinese
- 2019 • **A GIS-based high spatial resolution assessment of large-scale PV power generation potential and associated CO₂ emission reduction in China**
Applied Energy
- Qing Yang, Tianyue Huang, Saige Wang, Jiashuo Li, Shaoqing Dai, Sebastian Wright, Yuxuan Wang, Huaiwu Peng.
 - 247, 254-269. [doi:10.1016/j.apenergy.2019.04.005](https://doi.org/10.1016/j.apenergy.2019.04.005)
 - SCI, IF=11.0
- 2019 • **PM_{2.5} spatiotemporal evolution and drivers in the Yangtze River Delta between 2005 and 2015**
Atmosphere
- Guoliang Yun, Yuanrong He, Yuantong Jiang, Panfeng Dou, Shaoqing Dai.
 - 10(2), 55. [doi:10.3390/atmos10020055](https://doi.org/10.3390/atmos10020055)
 - SCI, IF=2.3
- 2018 • **Determining the Mechanisms that Influence the Surface Temperature of Urban Forest Canopies by Combining Remote Sensing Methods, Ground Observations, and Spatial Statistical Models**
Remote Sensing
- Shudi Zuo, Shaoqing Dai, Xiaodong Song, Chengdong Xu, Yilan Liao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.
 - 10(11), 1814. [doi:10.3390/rs10111814](https://doi.org/10.3390/rs10111814)
 - SCI, IF=4.1
- 2018 • **Analysis of Heavy Metal Sources in the Soil of Riverbanks Across an Urbanization Gradient**
International Journal of Environmental Research and Public Health
- Shudi Zuo, Shaoqing Dai, Yaying Li, Jianfeng Tang, Yin Ren.
 - 15(10), 2175. [doi:10.3390/ijerph15102175](https://doi.org/10.3390/ijerph15102175)
 - SCI, IF=4.6

- 2018 • **Influence of walking environment on robbery, snatch and theft crime in urban area, H city, China**
Scientia Geographica Sinica
- Shaoqing Dai, Huixian Jiang, Jiajia Li, Xian Su, Juan Wu, Yin Ren.
 - 38(8), 1235-1244. [doi:10.13249/j.cnki.sgs.2018.08.005](https://doi.org/10.13249/j.cnki.sgs.2018.08.005)
 - CSCD, in Chinese
- 2018 • **High-resolution mapping of direct CO₂ emissions and uncertainties at the urban scale**
Proceedings of Spatial Accuracy 2018.
- Shaoqing Dai, Shudi Zuo, Yin Ren.
 - 88-90.
 - EI
- 2018 • **Improving the prediction accuracy of forest aboveground biomass benchmark map by integrating machine learning and spatial statistics**
Proceedings of Spatial Accuracy 2018.
- SQ. Dai, Xiaoman Zheng, Shudi Zuo, Yin Ren.
 - 80-83
 - EI
- 2018 • **Individual and Interactive Influences of Anthropogenic and Ecological Factors on Forest PM_{2.5} Concentrations at an Urban Scale**
Remote Sensing
- Guoliang Yun, Shudi Zuo, Shaoqing Dai, Xiaodong. Song, Chengdong Xu, Yilan Liao, Peiqiang Zhao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.
 - 10(4), 521. [doi:10.3390/rs10040521](https://doi.org/10.3390/rs10040521)
 - SCI, IF=4.1
- 2018 • **Research on the effects of different policy tools on China's emissions reduction innovation**
China Population, Resources and Environment
- Qin Ye, Gang Zeng, Shaoqing Dai, Fenglong Wang.
 - 28(02), 115-122. [doi:10.12062/cpre.20170915](https://doi.org/10.12062/cpre.20170915)
 - CSCD, in Chinese
- 2017 • **Research on the effects of command-and-control and market-oriented policy tools on China's energy conservation and emissions reduction innovation**
Chinese Journal of Population Resources and Environment
- Qin Ye, Shaoqing Dai, Gang Zeng.
 - 16(1), 1-11. [doi:10.1080/10042857.2017.1418273](https://doi.org/10.1080/10042857.2017.1418273)
 - ESCI, IF=4.8

2017

• **Landscape-Scale Simulation Analysis of Waterlogging and Sponge City Planning for a Central Urban Area in Fuzhou City, China.**

International Low Impact Development Conference China 2016: LID Applications in Sponge City Projects.

•

Shaoqing Dai, Jiajia Li, Shudi Zuo, Yin Ren, Huixian Jiang.

•

251-260. [doi:10.1061/9780784481042.028](https://doi.org/10.1061/9780784481042.028)

•

EI

2016

• **The Analysis of Urban Spatial Development Pattern in Beijing Based on the Big Data of Government**

Geomatics World

• Miaoyi Li, Shaoqing Dai, Jingyuan Wang, Zhenjiang Shen.

• 23(3), 20-26. [doi:10.3969/j.issn.1672-1586.2016.03.004](https://doi.org/10.3969/j.issn.1672-1586.2016.03.004)

• JST



PUBLICATIONS-BOOK CHAPTER

2026

• **The evaluation of health effect of short-term exposure to PM2.5 during spring festival: a case study of 25 cities in the Yangtze River Delta**

In Applications of Medical Big Data and Artificial Intelligence(in Chinese) edited by Xifeng Wu, Guihua Zhuang, Yan Jin, Chaolong Wang, Chen Mao, Beijing: People's Medical Publishing House. ISBN:9787117387644.

• Shaoqing Dai, Huixian Jiang

2025

• **Healthcare service allocation and planning & Emerging fields in health geography**

In Health Geography(in Chinese) edited by Peng Jia, Beijing: China Higher Education Press. ISBN:978-7-04-063403-7

• Yuanyuan Shi, Shaoqing Dai, Peng Jia

2022

• **Assessing bicycle-sharing riding friendliness by fusing multi-source heterogeneous spatio-temporal big data**

Planning Decision Support Model Design Compilation 2021(in Chinese) edit by Beijing Municipal Institute of City Planning and Design, Beijing Chengyuan Digital Technology Co. Ltd., World Urban Planning Education Network, Beijing Planning Institute Hongdu Planning and Architectural Design Research Institute Co. Ltd., Beijing: China Architecture & Building Press. ISBN: 9787112270439

• Shaoqing Dai, Zhidong Chen, Jinghan Lei, Yanwen Wang

2019

Individual and Interactive Influences of Anthropogenic and Ecological Factors on Forest PM2.5 Concentrations at an Urban Scale

Advances in Quantitative Remote Sensing in China—In Memory of Prof. Xiaowen Li, edited by Shunlin Liang, Guangjian Yan, Jiancheng Shi, Remote Sensing, MDPI, 2019, 316–332. ISBN: 978-3-03897-276-1. <https://doi.org/10.3390/books978-3-03897-277-8>.

• Guoliang Yun, Shudi Zuo, Shaoqing Dai, Xiaodong Song, Chengdong Xu, Yilan Liao, Peiqiang Zhao, Weiyin Chang, Qi Chen, Yaying Li, Jianfeng Tang, Wang Man, Yin Ren.



PUBLICATIONS-PATENT

2020

Improving the accuracy of forest biomass estimate using source analysis and machine learning

CN111814397A

• Yin Ren, Xiaoman Zheng, Shaoqing Dai, Qi Chen, Shudi Zuo

2019

A biomass estimation method by fusing plot data and forest inventory data

CN110162872A

• Yin Ren, Shaoqing Dai, Shudi Zuo, Xiaoman Zheng.

2018

A prediction method of subtropical forest biomass

CN108959705A

• Qi Chen, Yin Ren, Xiaoman Zheng, Shudi Zuo, Shaoqing Dai



PUBLICATIONS-SCIENTIFIC AND TECHNOLOGICAL ACHIEVEMENTS

2025

'Environment + dual carbon goals' digital and intelligent platform development (Roadshow Signing Project).

Strait (Jinjiang) Technology Innovation Center Co., Ltd., 3rd National Postdoctoral Innovation and Entrepreneurship Competition.

• Shaoqing Dai, Wufan Zhao, Xinying Zhang, Zhixin Lin, Yuanyuan Shi

2023

Scaling up of carbon sequestration for Eucalyptus plantations based on ETKF-3DVAR hybrid data assimilation

Institute of Urban Environment, Chinese Academy of Sciences and Xiamen University of Technology, 90000890-2349, Xiamen Municipal Bureau of Science and Technology

• Yin Ren, Shudi Zuo, Lizhong Hua, Shaoqing Dai





PUBLICATIONS-SOFTWARE COPYRIGHT

2025

• **ISLE wearble device healthy data management platform. V 1.0**

• RN: 2025SR1194658

2019

• **A water saftety monitation system based on environmental IOT data and InVEST model. V1.0**

• RN: 2019SR0517519

2018

• **A water conservation service monitation system based on environmental IOT data**

• RN: 2018SR745897



PUBLICATIONS-OPEN DATASET & CODE

2024

• **Code and data to support 'Street view imagery for built environment auditing: a systematic review' (Version 5)**

Zenodo.

• Shaoqing Dai, Alfred Stein, Yuchen Li, Shujuan Yang, Peng Jia.

• [doi:10.5281/zenodo.10724305](https://doi.org/10.5281/zenodo.10724305)

2020

• **GISerDaiShaoqing/3dmodelFJNU 1.0 (Version 1.0)**

Zenodo.

• Shaoqing Dai

• [doi:10.5281/zenodo.4277977](https://doi.org/10.5281/zenodo.4277977)

2020

• **GISerDaiShaoqing/Urban-Carbon-Dioxide-sources-gridded-maps-and-itsdetemination-in-Jinjiang-city 0.6 (Version 0.6)**

Zenodo.

• Shaoqing Dai, Shudi Zuo, Yin Ren.

• [doi:10.5281/zenodo.3566072](https://doi.org/10.5281/zenodo.3566072)



RESEARCH PROJECTS

2028

|

2026

• **Research on key technologies for low-altitude UAV river-lake inspection and environmental dynamic perception and modeling**

Open Funding for Anhui Province Key Laboratory of Physical GeographicEnvironment (2025PGE006)

• PI

- 
- 2027
|
2025
- **The precise monitoring and analysis for residential environment risk factors and health effects among vulnerable populations**
Special Program for the Introduction of Overseas Postdoctoral Talents
• PI
- 2026
|
2024
- **Early Identification and Screening of Risk Factors for Chronic Complications in Diabetes Patients by Combining Traditional Chinese and Western Medicine**
Research Project of Collaborative Chronic Disease Management by Traditional Chinese and Western Medicine
• Investigator
- 2026
|
2023
- **Evaluation and prevention of health risks in the community home environment for the elderly and children, and development of appropriate products**
National Key Research and Development Plan(2023YFC3604704)
• Investigator
- 2025
|
2022
- **The mechanism of the impact of urban neighborhood-level walkability on child obesity**
National Natural Science Foundation of China(42271433)
• Investigator
- 2021
- **Investigating the Changes of Residents' Daily Behaviors, Lifestyles, and Built Environment During the COVID-19 Pandemic**
WeSpace Academic Support Plan in 2020: "Tsinghua-Tencent" Emerging Technologies Future Urban Space project
• PI
- 2020
|
2016
- **'Research on the key technology research and integration demonstration of urban agglomeration ecological security in the Yangtze River Delta**
National Key Research and Development Plan(2016YFC0502704)
• Investigator
- 2019
|
2018
- **The construction of Typhoon Disaster Assessment Index Based on GIS and Multi-source Remote Sensing Data in Fujian Province**
Open Fund of the Big Data Institute of Digital Natural Disaster Monitoring in Fujian (NDMBD2018001)
• Investigator
- 2018
|
2015
- **Scaling up of carbon sequestration for Eucalyptus plantations based on ETKF-3DVAR hybrid data assimilation**
National Natural Science Foundation of China(31470578)
• Investigator

2015
|
2014

The Optimization Model of High Education Staff's Refuge Space Under Earthquake Disaster

The Technological Innovation Plan of University Students, Fujian Normal University(cxsl-2014137)

• PI

PRESENTATIONS

2025

Sensing healthy city based on multimodal data and GeoAI

Aerospace Symposium on Data Technology and Data Curation, DTDC 2025
📍 Qingdao, China

2025

Toward health geography: from the COVID-19 pandemic to Health China

GISChat
📍 Online

2025

Sensing healthy city based on multimodal data and GeoAI

3rd Urban Agglomeration Green Development Conference
📍 Xiamen, China

2025

Sensing healthy city based on street view imagery and GeoAI

2nd National Conference of Information Geography
📍 Chengdu, China

2025

Application of night light remote sensing data in urban studies

Urban Party
📍 Online

2025

Sensing healthy city based on street view imagery and GeoAI

5th International Symposium on Spatial Lifecourse Health
📍 Wuhan, China

2025

Sensing healthy city based on street view imagery and GeoAI

Annual conference of The Geographical Society of Fujian
📍 Fuzhou, China

2025

Sensing healthy city based on street view imagery and GeoAI

3rd China Symposium on Urban Sustainable Development
📍 Wuhan, China

2025

The impact assessment of urban built environment on carbon-air-health nexus: a lifecourse perspective(invited)

3rd China Symposium on Urban Sustainable Development
📍 Wuhan, China

2024

Development of the frequently used built environmental auditing and simulation tools in health domain

4th International Symposium on Spatial Lifecourse Health
📍 Wuhan, China

2024

Development of the frequently used environmental exposome observation tools in health domain

2024 Health Geography Youth Scholars Forum
📍 Zhuhai, China

- 2024 • **Bridging environment and human health: from earth observation to human-center observation(invited)**
AI 4D City lab, Hong Kong University of Science and Technology
(Guangzhou)  Guangzhou, China
- 2024 • **Assessment of physical activity opportunities based on multi-source geospatial data**
9th Youth Geoscience Forum  Xiamen, China
- 2024 • **The frequently used built environmental auditing tools in the health domain**
1st National Remote Sensing and Geographic conference  Nanjing, China
- 2024 • **Health geography studies based on spatial lifecourse epidemiology framework**
Institute of Urban Environment, Chinese Academy of Sciences  Xiamen, China
- 2023 • **Seeking your interest and going ahead will contribute to success(invited)**
Xiamen No.1 High School of Fujian  Xiamen, China
- 2022 • **Do neighborhood boundaries matter for examining the associations between environmental factors and obesity(invited)**
The 29th International Conference on Geoinformatics in 2022 session 10-2
GIScience for Public Health  Beijing, China
- 2020 • **The Evaluation of Health Effect of Short-term Exposure to PM2.5 during Spring Festival: A Case Study of 25 Cities in the Yangtze River Delta(invited)**
The Workshop of 16th Spatial Behavior and Planning & Digitalization and Fine-Grained Governance  Xiamen, China
- 2020 • **What is geography? From coronavirus disease 2019 to Bei Dou Systems**
Tong An No.1 High School of Fujian  Xiamen, China
- 2018 • **The Scale Effect, Zoning Effect of Geographical Features, the Modified Areal Unit Problem and the Chanlleges of Spatial Stastistics**
11th Chinese R Language Conference  Beijing, China

- 2018

High-Resolution Mapping of Direct CO₂ Emissions and Uncertainties at the Urban Scale

13th International Symposium on Spatial Accuracy Assessment in Natural Resources and Environmental Sciences(Spatial Accuracy 2018)

Beijing, China
- 2017

The Environmental Effect of Urban Form on PM2.5: A Case Study of Beijing-Tianjin-Hebei Urban Agglomeration

9th Chinese Landscape Ecology workshop

Guangzhou, China
- 2017

The Assement Platform of Urban Forest Ecosystem Servcie Based on InVEST model and IOT(invited),

2rd urban environment long-term monitor and sustainable development workshop

Shanghai, China
- 2016

The Criminal Geographical Analysis about Walking Environment of Urban

The Workshop of 12th Spatial Behavior and Planning & Spatial-Temporal Behavior and Social Planning Research

Beijing, China
- 2016

Transportation Planning of Smart City on the Basis of Data Augmentation Design Under the Perspective of Humanism: A Case Study of Fuzhou's Cangshan District

10th annual conference of International Association of China Planning

Beijing, China

TEACHING EXPERIENCES

- Current
|
2025

Empowering GeoAI and Health Geography with R Programming

personalized cultivation week program course, School of Geographical Sciences, School of Carbon Neutrality Future Techonology, Fujian Normal University

 - 2 class hours
- Current
|
2024

Applications of Spatial Technology and AI in Health Geography

School of Resource and Environmental Science, Wuhan University

 - 4 class hours
- Current
|
2024

Applications of Remote Sensing technology in Public Health and Diseases Prevention

China CDC National Institute of Parasitic Diseases

 - 1 class hours

Current
|
2024

- **Applications of Spatial technology, Complex Science, and AI in Health Geography**
School of Resource and Environmental Science, Wuhan University
• 9 class hours



SELECTED POSTERS

2017

- **Quantifying the linear and nonlinear relations between the urban form fragmentation and the carbon emission distribution**
2017 AGU Fall Meeting  New Orleans, USA

2017

- **The relationship between the fragmented urban form landscape and the carbon emission distribution at different resolutions**
9th Lecture of Modern Ecology  Shanghai, China

2016

- **The Simulation Analysis of Waterlogging and the Sponge City Planning Control of Central Urabn Area in Fuzhou City**
2016 International Low Impact Development Conference  Beijing, China